Final

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2026-06-09

Set up

```
# load in packages
library(tidyverse)
library(here)
library(janitor)
library(DHARMa)
library(ggeffects)
# load in data
nest <- read_csv(here("data","nest_data_final.csv"))
eggs <- read_csv("C:/git/ENVS-193DS_homework-03/data/personal-data-project - Sheet1.csv")
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used a logistic regression. In part 2, they used a chi-square test.

b. Suggestions for rewriting

Part 1: These results indicate that the distance to the water's edge has an influence on whether on not the American Avocets will use a microhabitat.

Part 2: These results indicate that different species of birds such as the American Avocet, Forester's Tern, and Black-necked Stilt favor different habitat types.

c. Further analysis

Another variable mentioned in the paper is the American Avocet's access to islands for breeding and roosting and how far away the nesting island is. The distance to the nesting island is a continuous variable, measured in meters, and it was found that the Avocets were on average, 131 meters closer to the nesting island than expected. Access to these island is important for the rest and reproduction of American Avocets. If the birds are closer to the islands, they may choose to occupy that microhabitat over their other options since they seem to like them.

Problem 2. Data analysis

a. Clean and wrangle

```
# create new object from nest
nests_clean <- nest |>
  # select columns of interest (the columns I want to answer the research
  # question are height, ant species, nest occurrence)
  select(height_cm, ant_species, case_control) |>
  # filter for specific species
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  #
  mutate(
    # reorder the ant species
    ant_species = fct_relevel(ant_species, "AB", "RRB", "BBR"),
    # add a column for the scientific names
    scientific_name = recode_values(ant_species,
                                    "AB"  ~ "Crematogaster sjostedti",
                                    "RRB" ~ "Crematogaster mimosae",
                                    "BBR" ~ "Crematogaster nigriceps"))
# display structure of data frame
glimpse(nests_clean)
```

```
Rows: 270
Columns: 4
$ height_cm       <dbl> 391.5, 310.0, 513.0, 154.0, 324.5, 413.0, 167.0, 276.5~
$ ant_species     <fct> RRB, RRB, RRB, RRB, AB, RRB, BBR, RRB, AB, BBR, RRB, R~
$ case_control    <dbl> 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, ~
$ scientific_name <chr> "Crematogaster mimosae", "Crematogaster mimosae", "Cre~
```

```r
# display 10 random rows
nests_clean |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 4
   height_cm ant_species case_control scientific_name
       <dbl> <fct>              <dbl> <chr>
 1      302. RRB                    0 Crematogaster mimosae
 2      170  BBR                    0 Crematogaster nigriceps
 3      114. AB                     0 Crematogaster sjostedti
 4      168. RRB                    0 Crematogaster mimosae
 5      167  RRB                    0 Crematogaster mimosae
 6      155  BBR                    0 Crematogaster nigriceps
 7      207  RRB                    0 Crematogaster mimosae
 8      423  RRB                    0 Crematogaster mimosae
 9      199  RRB                    0 Crematogaster mimosae
10      310  RRB                    0 Crematogaster mimosae
```

b. Response variable

The 1s mean that the tree contained a nest and the 0s mean that the tree did not contain a nest.

c. Purpose of the study

Tree height: Tree height is a continuous variable because a tree can be any height measurement. As tree height increases, the probability of nest occurrence could increase because taller trees that are older may have thicker branches and more space to nest. This information is based on what I read in the results section in paragraphs 1 and 2 as well as the discussion section in paragraphs 1 and 2.

Acacia ant species: Acacia ant species is a categorical variable because it groups the trees by ant symbiont species. The tree with the more aggressive species of ants had a higher probability of having a bird nest because the ants are more likely to deter predators that might want to climb the tree to get to the bird nest. This is based on information I read in the results section in paragraphs 1 and 2.

d. Table of models

Model number	Tree height	Ant species	Interaction (Height × Species)	Predictor list
1	X	X		tree height and ant species
2	X	X	X	tree height, ant species, and their interaction

e. Run the models

```
# model 1: without interaction
model1 <- glm(
  case_control ~ height_cm + ant_species,
  data = nests_clean,
  family = binomial
)

# model 2: with interaction
model2 <- glm(
  case_control ~ height_cm * ant_species,
  data = nests_clean,
  family = binomial)
```

f. Select the best model

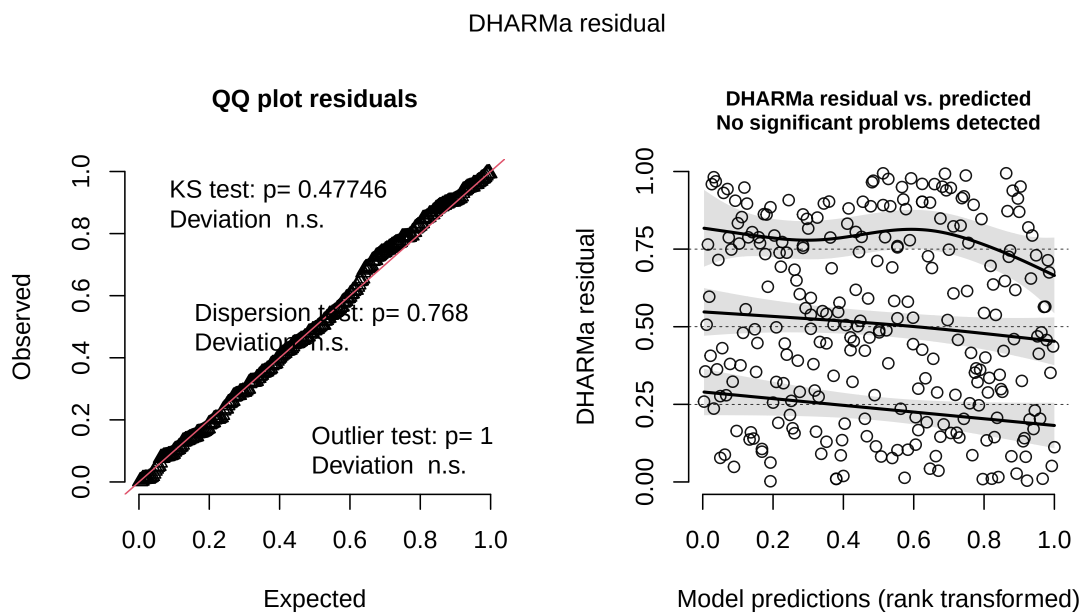
```
# compare models
AIC(model1, model2)
```

```
       df      AIC
model1  4 258.7430
model2  6 237.8042
```

The best model was model 2, the model with the interaction. This model includes the tree height, acacia ant species, and the interaction between them. This tells us if the relationship between tree height and the probability of a nest changes based on the ant species.

g. Check diagnostics

```
# simulate the residuals - use the selected model
simulated_res <- simulateResiduals(fittedModel = model2, n = 250)
# plot the diagnostic outputs
plot(simulated_res)
```



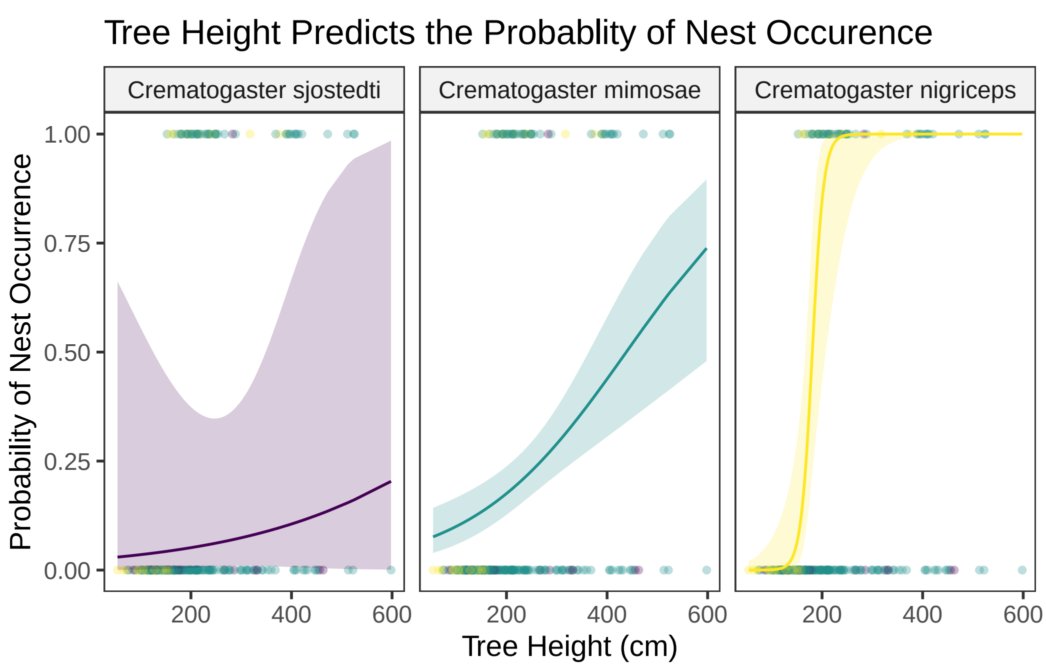
- Residuals are normally distributed, the points follow the red line (QQ plot)
- Residuals are homoscedastic, the points look random (DHARMa residuals vs. predicted)

h. Visualize model predictions

```
 # extract predicted probabilities along the range of tree heights for each ant species
 # [all] means look at every measured height
predictions <- ggpredict(model2, terms = c("height_cm [all]", "ant_species"))
# write out ant names so they can be used on the plot
ant_names <- c(
  "RRB"    = "Crematogaster mimosae",
  "BBR" = "Crematogaster nigriceps",
  "AB" = "Crematogaster sjostedti")
```

```
# base layer: ggplot
ggplot() +
  # 1st layer: geom_jitter to show the underlying data points
  geom_jitter(
    # use clean data frame
    data = nests_clean,
    # x-axis: tree height
    aes(x = height_cm,
        # y-axis: nest vs no nest
        y = case_control,
        # color by ant species
        color = ant_species),
    # allow slight horizontal jitter
    width = 0.1,
    # allow no vertical jitter
    height = 0,
    # make points mmore transparent
    alpha = 0.3,
    # make points smaller
    size = 0.9) +
  # 2nd layer: geom_line to show model predictions
  geom_line(
    # use predictions data frame
    data = predictions,
    # x-axis: x values
    aes(x = x,
        # y-axis: predicted values for nest
        y = predicted,
        # color by ant species
        color = group)) +
  # 3rd layer: geom_ribbon to show 95& confidence interval
  geom_ribbon(
    # use predictions data frame
    data = predictions,
    # x-axis: x
    aes(x = x,
        # ymin: lower bound 95% CI
        ymin = conf.low,
        # yman: upper bound 95% CI
        ymax = conf.high,
        # fill color by ant species
        fill = group),
```

```
    # make more transparent
    alpha = 0.2) +
  # separate by ant species
  facet_wrap(~ group,labeller = as_labeller(ant_names)) +
  # label axes and add title
  labs(
    x = "Tree Height (cm)",
    y = "Probability of Nest Occurrence",
    title = "Tree Height Predicts the Probablity of Nest Occurence") +
  # change colors
  scale_color_viridis_d() +
  scale_fill_viridis_d() +
  # change theme
  theme_bw() +
  theme(
    # take out gridlines
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    # take out legend
    legend.position = "none",
    # chhange fill background color
    strip.background = element_rect(fill = "grey95"))
```



i. Caption

Figure 1: Predicted probability of bird nest occurrence as a function of tree height across acacia ant host species. Shaded ribbons indicate the 95% confidence intervals around the predicted probabilities (lines) calculated from the chosen generalized linear model (model 2) in which there was an interaction between tree height and ant species. Raw data points (jittered horizontally along the binary 0 and 1 outcomes) represent observed nest absences and presences sampled across varying tree heights. Data adapted from Palmer et al. (2017), available via the Dryad Digital Repository.

j. Calculate model predictions

```r
# create new data frame for 600 cm
prediction_600cm <- data.frame(
  # define height
  height_cm = c(600, 600, 600),
  # for each species
  ant_species = c("RRB", "BBR", "AB")
)
# calculate predicted probabilities using model2 and the new data frame
predicted_probabilities <- predict(model2,
                                   newdata = prediction_600cm,
                                   # type = "response" converts the internal
                                   # log-odds into a 0-1 probability scale
                                   type = "response")
# create summary table
results_table <- data.frame(prediction_600cm,
                            Predicted_Probability = predicted_probabilities)
# show table
print(results_table)
```

```
  height_cm ant_species Predicted_Probability
1       600         RRB             0.7407857
2       600         BBR             1.0000000
3       600          AB             0.2048017
```

k. Interpret results

Based on the model 2, tree height and acacia ant species are both critical predictors determining the probability of bird nest occurrence. As visualized in Figure 1, there is a strong, positive relationship between tree height and nest occupancy, though the initial probability is different depending on the ant species that lives in the tree. For the tall trees (at 600 cm), the model predicts a high probability of nest occupancy when hosted by Crematogaster nigriceps (close to 100%) and a high probability within Crematogaster mimosae trees (74%), whereas the probability of a nest in trees occupied by Crematogaster sjostedti is low (20%). Biologically, these trends could be explained by variations in ant behavior and host-tree modification: C. nigriceps aggressively prunes canopy buds to reduce visibility and structural access for predators, which adds protection for nest occupants. Both C. mimosae and C. nigriceps strongly defend their territories against elephants, which protects the tree and reduces distrubances for nesters. Whereas C. sjostedti participates in wood boring behaviors that can degrade the tree, and make it a less secure nesting site for birds.

Problem 3.

a. Comparing visualizations

My exploratory visualizations from homework 2 have so few data points that it was hard to see a trend. I didn't really have a good idea or guess about how my variables would relate to each other. My early visualizations just (somewhat effectively) show points and data that I collected. My affective visualization represents the connection that I built with my data and its collection over the course of the quarter.

Throughout my project, I wanted to be able to predict the number of eggs I ate. It was not until we learned about linear models that I realized what specific question I wanted to answer and why.

All of my visualizations end up showing the same negative relationship between number of eggs consumed and servings of meat eaten. From the first time I plotted those two variables up until I made my affective visualization, I thought that was the most important takeaway to highlight from my data collection.

During week 9, I was encouraged to show some reference to yogurt consumption, another important variable I collected data on. I didn't know how to draw yogurt, but I tried adding spoons to the border of my visualization. My yogurt consumption was very consistent no matter how other variables changed so I thought a border would be appropriate. In the end, I took it out again because I thought it was really weird and distracted from the stars of the show. I really wanted to show chicken increasing as eggs decreased and that was my main goal.

b. Designing and analysis

I would like to do a simple linear regression. This will tell me if there is a directional, linear relationship between my egg consumption and my meat consumption, which is what I've wanting to know all along. It is appropriate for my variables because eggs and meat are both numeric variables that I can use to make a trend line. After doing this, I will be able to determine if how increases in meat consumption will affect my eggs consumption.

c. Check assumptions and run tests

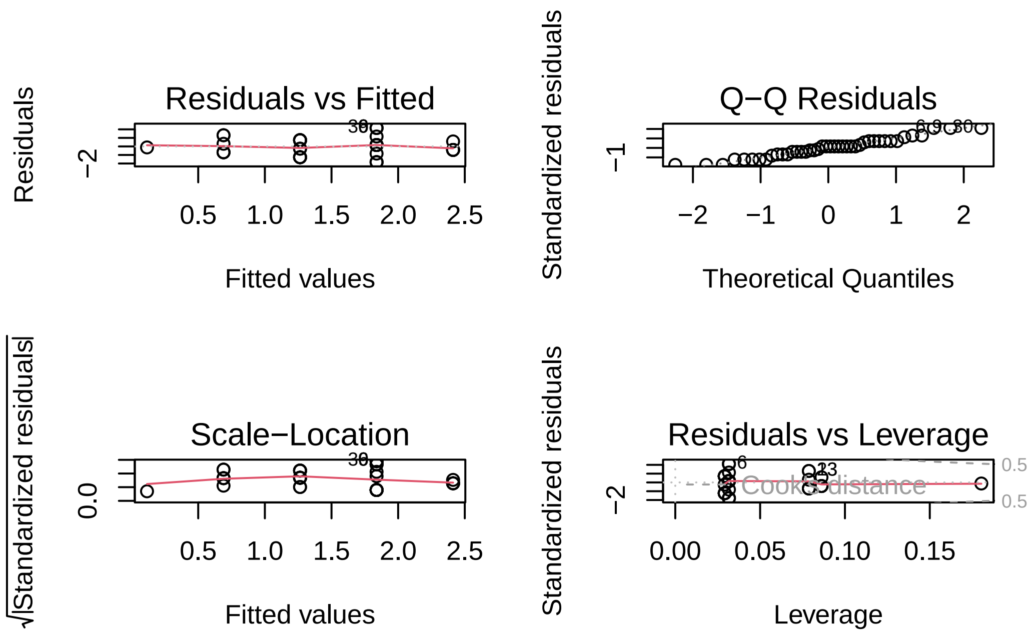
```r
# create new object from egg_data
egg_data_clean <- eggs |>
# clean names
clean_names() |>
# remove any row where all values are NA
remove_empty(which = "rows") |>
# remove the empty (1st) column that was all NA
select(-1) |>
# change names from 1, 2, 3, etc.
set_names("date",
          "number_eggs_eaten",
          "servings_yogurt_eaten",
          "breakfast_at_school",
          "servings_meat_eaten", "school_day") |>
# remove the first row (old spreadsheet cloumn titles)
slice(-1) |>
# convert the columns from text to numbers
mutate(number_eggs_eaten = as.numeric(number_eggs_eaten),
       servings_yogurt_eaten = as.numeric(servings_yogurt_eaten),
       servings_meat_eaten = as.numeric(servings_meat_eaten))
```

```r
# create new object to store linear model
# formula: lm(dependent ~ independent, data = data frame)
egg_model <- lm(number_eggs_eaten ~ servings_meat_eaten,
data = egg_data_clean)
# show summary of linear model
summary(egg_model)
```

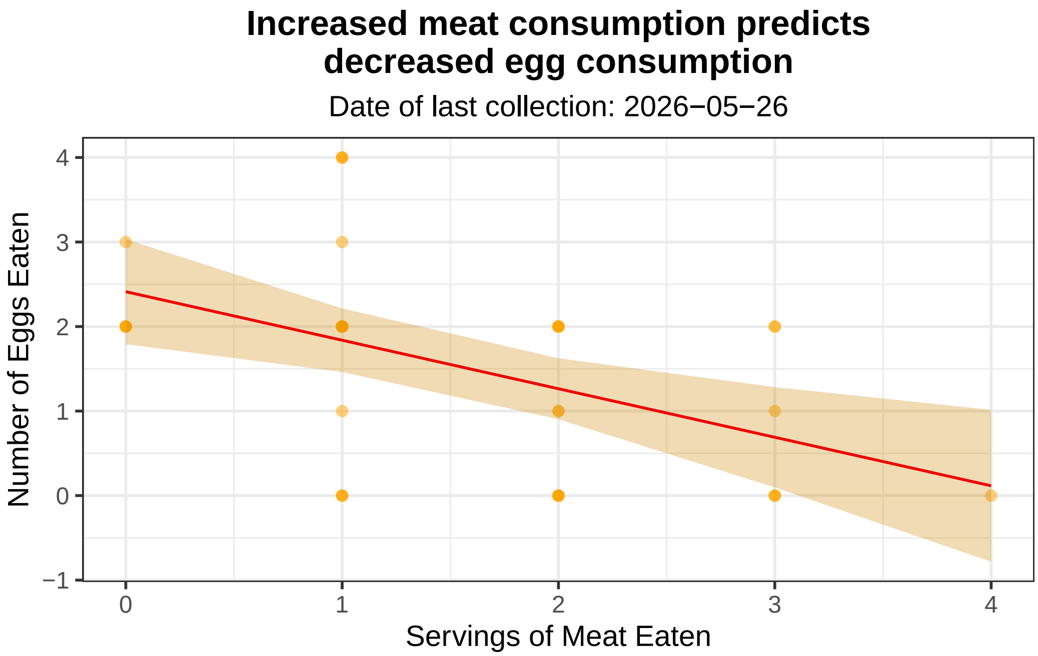
```
Call:
```

```
lm(formula = number_eggs_eaten ~ servings_meat_eaten, data = egg_data_clean)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.8382 -0.6900  0.1618  0.7359  2.1618 

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)           2.4123     0.3072   7.851 1.26e-09 ***
servings_meat_eaten  -0.5741     0.1689  -3.399  0.00154 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.047 on 40 degrees of freedom
Multiple R-squared:  0.2241,	Adjusted R-squared:  0.2047 
F-statistic: 11.55 on 1 and 40 DF,  p-value: 0.001543
```

```r
# create 2x2 grid to show diagnostic plot
par(mfrow = c(2,2))
# show diagnostic plots
plot(egg_model)
```



```r
# create new object to store predictions of the model
egg_predict <- ggpredict(egg_model, terms = "servings_meat_eaten")
# base layer: ggplot
ggplot(data = egg_predict, # use predictions data frame
aes(x = x, # x-axis: x
y = predicted # y-axis: predicted
)) +
# 1st layer: geom_point to show raw data points
geom_point(data = egg_data_clean,
aes(x = servings_meat_eaten,
y = number_eggs_eaten), #
color = "orange", # point color = dark green
alpha = 0.5) + # make points slightly transparent
# 2nd layer: ribbon to show 95% CI
geom_ribbon(aes(ymin = conf.low, # lower bound of 95% CI
ymax = conf.high), # upper bound of 95% CI
fill = "orange3",
alpha = 0.3) + # make ribbon transparent
# 3rd layer: line representing model predictions
geom_line(color = "red2") +
# relabel x and y axes and add title
labs(x = "Servings of Meat Eaten",
y = "Number of Eggs Eaten",
title = "Increased meat consumption predicts
decreased egg consumption",
subtitle = "Date of last collection: 2026-05-26") +
# change theme
theme_bw() +
# center title and make text bold
theme(plot.title = element_text(hjust = 0.5,
face = "bold"),
# center subtitle
plot.subtitle = element_text(hjust = 0.5))
```


Increased meat consumption predicts
decreased egg consumption
Date of last collection: 2026−05−26
Number of Eggs Eaten
4
3
2
1
0
−1
0
1
2
3
4
Servings of Meat Eaten